Review of 2-Photon Microscopy in vivo Calcium Imaging Studies

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Spatiotemporal properties of cortical excitatory and inhibitory neuron activation by sustained and bursting electrical microstimulation

Intracortical microstimulation (ICMS) is used to evoke artificial sensations, which can be recorded through two-photon microscopy in transgenic mice. High-frequency ICMS causes neuron activity to fade quicker compared to lower frequencies. Some neurons respond more to low frequencies, high frequencies, or burst profiles. Decreasing responsiveness to ICMS over time is a constraining factor for functional applications. Two major hypotheses explain this: First, ICMS increases metabolic consumption faster than blood flow can replenish oxygen, leading to decreased measured activity over time. Second, inhibitory neurons reduce neural activity during ICMS.

Mice were genetically modified to express red fluorescence (AAV-FLEX-tdTomato) in inhibitory neurons and green fluorescence (AAV-Syn-GCaMP7b) in all neurons. Due to calcium indicator leakage, neurons were put into three neuron categories: excitatory (54%), inhibitory (16%), and unknown (30%)Inhibitory neurons showed a greater increase in ICMS-evoked fluorescent intensity as well as a slower time-to-peak and return-to-baseline than excitatory neurons. Neurons that responded to one ICMS pattern were more likely to respond to the other three patterns, suggesting that the pattern of stimulation does not significantly change the population of neurons recruited. Overall, inhibitory neurons were more strongly activated by ICMS than excitatory neurons.

Regarding stimulation patterns, 10-Hz uniform, 10-Hz burst, Theta-burst stimulation (TBS), and 100-Hz uniform stimulation were utilized. The strongest activation visualized by neuron fluorescent intensity during the first few seconds across all subpopulations was evoked by 100-Hz ICMS. Low-frequency patterns sustained activation over longer durations of ICMS, resulting in stronger activation during the stabilization period. Results indicate that 63% of inhibitory neurons prefer TBS compared to 35% of excitatory neurons. Conversely, 34% of excitatory neurons prefer 10-Hz burst compared to 14% of inhibitory neurons.

Decreases in activity during ICMS correlated with depression post-ICMS. Previous studies found that neurons can decrease in fluorescent intensity during ICMS and remain below baseline after ICMS ends, indicating synaptic plasticity or neuronal exhaustion.
Higher-frequency stimulation patterns led to greater post-ICMS depression compared to lower frequencies. TBS did not induce post-ICMS depression in inhibitory neurons and led to the least post-ICMS depression in excitatory neurons. Inhibitory neurons had a greater proportion of post-ICMS elevated activity compared to excitatory neurons. Neurons classified as decreasing during ICMS were more likely to be depressed post-ICMS, while neurons classified as increasing during ICMS were very likely to be elevated post-ICMS.

When comparing how neuron activation changed across distance, both inhibitory and excitatory neurons were less active further away from the electrode; however, inhibitory neurons remained more active than excitatory neurons. This indicates that indirect recruitment more strongly activates inhibitory neurons than excitatory neurons. Neurons close to the electrode preferred the 10-Hz burst pattern, whereas neurons further away preferred TBS.

Activity-dependent recruitment of inhibition and excitation in the awake mammalian cortex during electrical stimulation

Functional activity within circuits at the time of stimulation modulates evoked activity patterns. Characterizing the effects of ongoing neural activity on stimulation-evoked activity patterns is necessary to understand how stimulation is integrated into neural circuits in awake states and affects the perception and behavior of the animal. ICMS at high frequency becomes complex as stimulated pulses interact with activity patterns initiated by prior pulses leading to variable temporal responses. In general, low stimulation frequencies are excitatory and correlate with the amplitude and spatial extent of neural responses, while high stimulation frequencies are suppressive, promoting release of inhibitory neurotransmitters. The goals of this study were to record activation of inhibitory neurons to electrical stimulation and determine how stimulation-evoked neural activity is integrated into ongoing neural circuits.

The visual cortex was recorded with pan-neuronal calcium indicator GCaMP6s, while inhibitory neurons were labeled with red fluorescence by crossing Tdtomato reporter line Ai14 with Gad2-ires-cre driver line, similar to the last study. Increasing stimulation amplitude increases the number of recruited neurons for both excitatory and inhibitory neurons. Results, indicate that with increasing stimulation amplitude indirect excitation increases, while direct excitation decreases. Additionally, because inhibitory neurons have denser local axonal arborizations within cortical layers they are preferentially activated at low stimulation amplitudes, this leads to suppression in neural activity.

Consistent with prior work, as stimulation amplitude increased, the overall 3D volume of space containing modulated cell bodies didn't change significantly for excitatory neurons, and only increased after moderate amplitude increase for inhibitory neurons.

Results found that stimulation amplitude and pre-stimulus activity affect stimulation-evoked responses with a positive average correlation, indicating that at higher stimulation amplitudes generally lead to larger evoked responses in all neurons. Conversely, pre-stimulus neural activity was negatively correlated with stimulation-evoked responses, suggesting that stimulation tended to suppress ongoing neural activity.

Overall, electrical stimulation is capable of modulating neural activity patterns in the brain. Activating a single putative inhibitory neuron elicits a stronger but more variable response than activating a single putative excitatory neuron. Thus, targeting excitatory neurons alone or in combination with a stimulation pattern that activates SST inhibitory neurons could improve artificial sensation. Additionally, increasing stimulation amplitude ot a certain extent could improve stability of evoked responses by reducing their dependence on ongoing activity aptters in the brain, which could help learning and integration of artificial sensory signals into natural sensorimotor function.

Learning induces activation-mechanism–dependent neural plasticity in an intracortical microstimulation task

Electrical microstimulation across weeks of learning alters neural responses to ICMS over time. Conceptually distinct from previous studies, these results indicate that ICMS-evoked plasticity is not a uniform change, but depends heavily on both behavioral context and underlying activation mechanisms.

The study utilized ultraflexible nanoelectronic threads (NETs) for both stimulation and simultaneous spike recording. Within each animal, three NET contacts were designated as stimulation sites and used consistently across all sessions. Crucially, a control group received chronic ICMS without the behavioral task, demonstrating that these neural adaptations result specifically from behavioral learning rather than repeated electrical stimulation alone.

Behavioral learning enhances ICMS-evoked population activation over time, as detection thresholds decrease throughout training. Additionally, at a fixed current, the 3D volumetric size of activation density increased over time. However, when researchers adjusted the current downward to match the animal's newly improved detection threshold, the number and density of activated neurons remained stable. This suggests that a consistent neural ensemble is required to maintain a behavioral response, even as the activation threshold drops.

By analyzing neural activity at millisecond resolution between stimulation pulses, researchers discovered two distinct response types. This addressed the central question of whether threshold changes reflect altered excitability or new population recruitment, demonstrating that it is actually a combination of both:

Pulse-Locked Neurons: Spikes align precisely with individual pulses, indicating direct activation (with delays increasing at greater distances from the stimulation site). Over learning, these neurons—specifically excitatory pyramidal cells, rather than fast-spiking interneurons—exhibited increased excitability and reduced latency.

Non-Pulse-Locked Neurons: Spikes are distributed over time, indicating indirect activation through broader network connections. Instead of changing excitability, these neurons expanded in number and predicted behavioral outcomes. As training progressed, non-pulse-locked neurons showed a substantial divergence in firing rates between "hit" and "miss" trials before movement initiation, linking artificial sensory input directly to behavioral relevance.

Fast online deconvolution of calcium imaging data

Extracting individual neuron spiking activity from fluorescent calcium imaging data is a computationally intensive process. Current state-of-the-art convex solvers rely on interior point methods that are too slow for real-time analysis of large-scale datasets, such as whole-brain imaging, and cannot process data sequentially. This severely limits their utility for closed-loop experiments that require on-the-fly analysis to guide experimental steps or control feedback.

To address this limitation, the authors developed the Online Active Set method to Infer Spikes (OASIS). This algorithm approaches activity extraction as a sparse non-negative deconvolution problem. By generalizing the pool adjacent violators algorithm (PAVA) for isotonic regression, OASIS processes time-series fluorescence data sequentially from beginning to end. This inherently online structure provides linear-time computational complexity and allows the algorithm to be warm-started with previous solutions, drastically reducing the time required to optimize model hyperparameters.

Results indicate that OASIS is at least an order of magnitude faster than leading convex solvers while maintaining identical high-quality spike inference. The algorithm's speed and memory efficiency enable the real-time, simultaneous deconvolution of neural traces—such as those generated in whole-brain larval zebrafish imaging—using only a standard laptop. Additionally, the researchers demonstrated that modifying the algorithm to impose a hard constraint on minimum spike size (a penalty) can further improve the precision of the neural activity inference by eliminating small, noise-driven values.

Robustness of Spike Deconvolution for Neuronal Calcium Imaging

Inferring exact spike times from calcium imaging is computationally challenging. While previous studies suggested complex supervised learning algorithms like Convolutional Neural Networks (CNNs) were most accurate, researchers found these methods overfit training data. When applied to novel datasets, CNNs generalize poorly and introduce artifactual structures, such as spurious auto-correlations, which could lead to erroneous scientific conclusions.

Using large-scale datasets and a novel benchmark that does not require ground-truth electrophysiology, the study demonstrated that a simple, unsupervised method—non-negative deconvolution (NND)—outperformed all other algorithms on out-of-sample data. Unconstrained NND proved highly robust to variations in the calcium indicator's assumed timescale. Standard exponential decay values from existing literature outperformed complex or automatically fitted kernels, and adding L0 or L1 regularization penalties offered no performance benefit.

Furthermore, NND is approximately two orders of magnitude faster than CNN-based approaches, running efficiently on standard CPUs rather than high-end GPUs. Due to its superior out-of-sample accuracy, speed, and resistance to overfitting, the authors recommend unconstrained NND for extracting spike times. These principles are integrated directly into large-scale pipelines like Suite2p, which relies on unconstrained NND via the fast OASIS algorithm to ensure unbiased spike inference.

Extracting large-scale neural activity with Suite2p

Optical recording methods such as two-photon microscopy enable the tracking of massive neural populations, with commercial microscopes now capable of recording over 100,000 neurons simultaneously. Processing the immense amount of raw data generated by these experiments requires highly efficient algorithms to extract interpretable neural timecourses. To address this, researchers developed Suite2p, a GPU-accelerated computational pipeline that performs motion correction, cell detection, activity extraction, and quality control in a fraction of the original acquisition time.

Brain motion in awake, in vivo recordings requires precise non-rigid correction due to sequential pixel-by-pixel acquisition. Suite2p initiates motion correction by iteratively refining a reference image from a subset of highly correlated frames to prevent blurring. It then applies global rigid correction using phase-correlation—which emphasizes edge features over low frequencies—followed by local non-rigid correction calculated in overlapping blocks with adaptive smoothing based on signal-to-noise ratio (SNR). When benchmarked, Suite2p's registration outperformed both Caiman and Fiola, running over five times faster and avoiding the gradual temporal misalignment observed in Caiman's online template updating.

For Region of Interest (ROI) detection, standard matrix decomposition approaches often falsely identify neuropil or motion artifacts as cells. Suite2p’s detection algorithm isolates cellular compartments by their transient functional excursions above baseline, functioning as a matrix decomposition with strict L0 sparsity constraints. The algorithm bins and high-pass filters the registered movie, convolving square templates of varying sizes to iteratively find and subtract regions of high variance. When tested against hybrid ground-truth data, this detection method found more cells and produced fewer false positives and false negatives than Caiman's CNMF algorithm, while running significantly faster. Alternative algorithms like Sourcery (for low SNR) and Cellpose (for anatomical segmentation) are also included in the pipeline.

Following detection, neural activity traces are extracted by averaging pixels, conservatively excluding regions where ROIs overlap to prevent signal contamination. A surrounding neuropil signal is subtracted before applying the OASIS algorithm with an AR-1 model for high-speed spike deconvolution. The pipeline features a graphical user interface (GUI) that calculates and displays quality control metrics, allowing users to assess registration stability and verify that cells are extracted uniformly. Due to its computational efficiency, Suite2p integrates directly with acquisition software like ScanImage for real-time Z-drift estimation, allowing large-scale datasets to be fully processed within 20 minutes of recording completion.

Low-threshold, high-resolution, chronically stable intracortical microstimulation by ultraflexible electrodes

Ultraflexible stim-Nanoelectronic Threads (StimNETs) demonstrate low activation thresholds, high spatial resolution, and chronically stable intracortical microstimulation (ICMS) during in vivo two-photon imaging. The study revealed that StimNETs remain seamlessly integrated with nervous tissue throughout chronic stimulation periods at low currents of 2 μA. Furthermore, StimNETs induced no neuronal degeneration or glial scarring, suggesting that tissue-integrated electrodes provide a robust platform for artificial sensation via ICMS.

Based on fundamental biophysical principles, implants that prevent glial scarring minimize the physical separation between electrode contacts and target neurons. This reduces activation thresholds, decreases the number of neurons activated at threshold, improves stimulation focality, and enables high-resolution, chronically stable neuromodulation. Previous work demonstrated that Nanoelectronic Threads (NETs) with a total thickness of 1 μm form an intimate tissue-electrode interface during chronic implantation, maintaining an intact blood-brain barrier, subcellular tissue stability for weeks, and an absence of glial scarring. Through iterative design optimization, ultraflexible StimNETs were engineered to minimize electrical crosstalk between nearby trace lines, enhance charge storage capacity, and protect against potential cracks on vertical walls using offset vias and contacts. To reduce the risk of delamination, StimNETs incorporate a microfabricated cap ring for mechanical reinforcement, bringing total shank thickness to 1.3 μm.

StimNETs can safely deliver stimulation currents up to 50 μA while remaining strictly within the water oxidation/reduction window (-0.6 V to 0.8 V), avoiding electrolysis and biological tissue damage.

The study utilized Thy1-GCaMP6s transgenic mice to selectively image excitatory neurons. Two-photon microscopy z-stacks were collected to generate 3D volumetric scans of cortical Layer 2/3 (~100–300 μm depth in somatosensory cortex). By calculating fluorescence changes between stimulation and baseline conditions across multiple trials, researchers identified consistently activated voxels and established a probability threshold to isolate neural ensembles driven specifically by ICMS.

PatchWarp: Corrections of non-uniform image distortions in two-photon calcium imaging data by patchwork affine transformations

Two-photon microscopy is heavily relied upon in neuroscience to record the activity of large neuronal populations at a high spatial resolution in behaving animals. By utilizing fluorescent calcium indicators, these optical systems track intracellular calcium increases to infer neuronal spiking. However, in vivo imaging is frequently corrupted by motion artifacts that must be surgically removed prior to data extraction. While fast, uniform translations are easily corrected via rigid alignment algorithms, extended imaging sessions suffer from slow, non-uniform spatial distortions. These complex spatial warps result from tissue mechanics, blood vessel dilation, metabolic activity, and mechanical system drift. When standard non-rigid correction algorithms are applied, they often fail to mitigate the warp, causing targeted neural structures to drift out of established regions of interest over time.

To robustly correct these slow image distortions, researchers developed PatchWarp, a novel two-step registration algorithm. The first stage of the pipeline corrects rigid motion artifacts by iteratively re-estimating a baseline template image. The crucial second stage addresses the non-uniform distortions by splitting the field of view into small overlapping patches, identifying the best affine transformation matrix for each subfield, and stitching them together like a patchwork to rebuild a stabilized, distortion-free image.

PatchWarp demonstrates significantly higher accuracy than existing non-rigid algorithms, proving highly effective at handling challenging datasets collected through glass windows or GRIN lenses. Beyond improving within-session accuracy, PatchWarp offers a fully automated mechanism for registering images from entirely different imaging sessions, making it highly valuable for longitudinal studies tracking the same neurons across weeks. The pipeline is openly available as a MATLAB package, providing a robust solution to a major limitation in long-term neuronal calcium imaging.

Dynamic amplitude modulation of microstimulation evokes biomimetic onset and offset transients and reduces depression of evoked calcium responses in sensory cortices

Intracortical microstimulation (ICMS) is used in bidirectional brain-computer interfaces (BCIs) to restore tactile and visual sensations, yet continuous stimulation with fixed parameters often causes evoked percepts to adapt and completely extinguish within a minute. While normal sensory input naturally evokes strong transient neural activity at stimulus onset and offset with modest firing in between, traditional linear ICMS fails to replicate this pattern and instead drives depression during continuous stimulation (DCS) and long-term stimulation-induced depression of neuronal excitability (SIDNE). To determine whether biomimetic encoding can overcome these barriers, this study evaluated how dynamically modulating ICMS amplitude and frequency affects the spatiotemporal activation and depression of cortical neurons.

The study utilized two-photon microscopy in anesthetized Thy1-GCaMP6s transgenic mice to record calcium responses from Layer 2/3 excitatory neurons in the visual and somatosensory cortices. Stimulation was delivered via 16-channel Michigan probes implanted at a 30° angle, testing both Short (1 s on, 4 s off, charge-balanced at 360 nC) and Long (30 s on, 15 s off) stimulus trains, alongside an elevated hold-amplitude condition (Long+). Four encoding profiles were compared: Fixed (constant amplitude and frequency), DynAmp (high-amplitude 200 ms onset/offset transients with a 25% reduced-amplitude hold phase), DynFreq (high-frequency 200 Hz transients with a 50 Hz hold phase), and DynBoth (simultaneous modulation of both amplitude and frequency).

Dynamic amplitude modulation (DynAmp and DynBoth) evoked distinct onset and offset transients across 100% of recruited neurons, achieved faster times-to-peak, and significantly lowered total charge injection during extended stimulation. By reducing stimulation amplitude during the prolonged hold phase, DynAmp and DynBoth trains recruited substantially fewer neurons (only 2–4% active during Long holds) at lower intensities, allowing the broader neural population to recover and maintain excitability for the offset transient. Conversely, DynFreq and Fixed trains maintained high recruitment throughout the hold phase, driving rapid and severe DCS that obscured the offset response. Dynamic frequency modulation alone only evoked biomimetic onset and offset transients in a small subpopulation of neurons (1–5%). However, lowering the stimulation frequency during the hold phase still reduced DCS in continuously recruited neurons compared to Fixed high-frequency trains, though DynFreq unexpectedly increased cross-trial depression (SIDNE) due to higher overall activation intensities.

Across all stimulation profiles, individual neurons exhibited heterogeneous responses strongly tied to their distance from the electrode—except when spared from continuous recruitment by dynamic amplitude modulation. During Fixed and DynFreq stimulation, neurons farther from the electrode (>200–400 μm) depressed much more rapidly than proximal neurons (<50 μm), which remained stable or facilitated. Notably, the small subpopulation of neurons capable of tracking dynamic frequency changes was located significantly farther from the electrode, suggesting that frequency modulation relies on indirect synaptic integration and local inhibitory network dynamics rather than direct entrainment. Furthermore, separating 30-second

DCS into two temporal phases revealed that initial rapid depression (the first 10 s) is heavily parameter-dependent, whereas slow depression (the final 20 s) follows a consistent logarithmic decay determined solely by the starting activation intensity at the beginning of the hold phase.

To evaluate the functional utility of these profiles for sensory encoding, an ideal observer analysis using Bayesian classifiers was trained on the evoked population calcium traces. Biomimetic encoding with DynAmp and DynBoth significantly accelerated the detection of stimulus onset (by ~0.03 s) and drastically reduced the latency for offset detection (by ~1.33 s) while generating fewer false alarms compared to linear encoding with Fixed and DynFreq trains. All train types maintained high accuracy (>75%) in discriminating between standard (Long) and elevated (Long+) hold amplitudes, confirming that biomimetic amplitude modulation preserves intensity discrimination while enhancing stimulus salience and maintaining long-term neural excitability.

3D Volumetric z-stack imaging utilizing Neurolabware Scanbox

Easy z-stack with the ETL

In vivo multiplane two-photon imaging has traditionally relied on motorized stages or piezo objective collars, whose physical inertia, acoustic vibration, and mechanical oscillations risk perturbing animal behavior and destabilizing delicate invasive implants like StimNETs or Michigan probes. The Easy z-stack with the ETL feature in Neurolabware Scanbox replaces physical movement with optical focal shifting by integrating an Optotune Electrically Tunable Lens (ETL) into the excitation beam path. By altering the curvature of an optical polymer fluid container via an electromagnetic voice coil, the system shifts the focal plane in 10–15 ms without moving the objective or sample.

Scanbox streamlines this setup within its Volumetric Scanning GUI panel. An experimenter navigates live to the upper and lower anatomical boundaries of interest, clicks "Top" and "Bottom" to latch the coordinates, and inputs the desired axial step size (micrometers) alongside frame-averaging counts. The software linearizes the voice-coil current against physical axial displacement and compiles a lookup table that uploads directly to the Scanbox acquisition tower FPGA. The hardware then cycles autonomously through the pre-programmed focal planes during scanning, providing rapid, vibration-free volumetric stacks to track 3D neural ensembles and depth-dependent intracortical microstimulation (ICMS) recruitment across cortical layers.

A depth stabilization plugin

While rigid and non-rigid in-plane registration pipelines like Suite2p and PatchWarp readily correct lateral (XY) motion, axial (Z) displacement presents a major confound in awake recordings. Because high-NA two-photon optical sections are only 1.5–3 micrometers thick, slow thermal drift, brain relaxation, and animal movement displace somata out of the focal plane, generating artificial step changes in raw fluorescence that mimic or obscure genuine calcium transients. To solve this, Scanbox implemented A depth stabilization plugin, a closed-loop software solution operating inside the synchronous sbxplugin architecture without lagging the primary image acquisition thread.

The plugin functions by evaluating live frames against an a priori high-resolution reference Z-stack (plus or minus 10–30 micrometers around the target plane). Using 2D fast Fourier transform (FFT) phase cross-correlation, it evaluates structural congruence across labeled somata, baseline neuropil, and vessel shadows. The reference slice producing the highest correlation peak identifies the instantaneous depth, yielding an error value (Delta Z) that is converted into a physical translation command sent to the motorized Knobby stage or ETL driver. This closed-loop feedback stabilizes the imaging depth in real time, preventing false fluorescence fluctuations during prolonged behavioral and ICMS experiments.

Automatic gain control of laser power

Cortical tissue acts as an inhomogeneous scattering medium where two-photon excitation light decays exponentially with depth according to the Beer-Lambert law ($P(z) = P_0 e^{(-z / l_s)}$, with scattering mean free path $l_s \approx 150–200$ μm at 920 nm). Consequently, static laser power causes severe overexposure and photobleaching in superficial layers or collapses the signal-to-noise ratio (SNR) in deeper layers. Scanbox addresses this through Automatic gain control of laser power (integrated into its broader Z-stack architecture), dynamically scaling excitation intensity as a function of imaging depth.

Optical attenuation is controlled via an Electro-Optic Modulator (EOM / Pockels cell) in the excitation path. Because Pockels cell transmission is non-linear ($T(V) = \sin^2(\pi V / (2 V_\pi))$), Scanbox calibrates and linearizes analog control voltages to enforce precise, predictable power delivery. During volumetric Z-stack execution, the system dynamically recalculates the required excitation power based on user-defined boundary percentages or an exponential attenuation constant (λ_z) and updates the analog control voltage synchronously with each axial step. This maintains uniform photon density, balanced baseline fluorescence (F_0), and consistent detection sensitivity across cortical layers without inducing superficial tissue heating or phototoxicity.

Confounding Variables and Methodological Artifacts in Intracortical Microstimulation (ICMS) Calcium Imaging

Confounding Variable	Primary Mechanism	Visualized ΔF/F0 Artifact	Spatial & Temporal Scale	Primary Literature
ATP-Depleted Ca2+ Pump Failure	High-frequency/amplitude ICMS exhausts local O2/ATP; PMCA, SERCA, and Na+/K+-ATPase/NCX stall	False steady-state excitation (high ΔF/F0 persists while actual spiking drops or ceases)	<50–80 μm from tip; onset within 10–20 s of continuous train	Hughes & Kozai (2023); Hughes et al. (2025)
Extracellular K+ Buildup & Conduction Block	Excess extracellular K+ and intracellular Na+ cause depolarization block or virtual anode block	Sudden drop-out ("rapidly adapting" somata) or long-term threshold elevation (SIDNE)	<100 μm during 50–250 Hz trains; SIDNE lasts 2–17 days	Eles & Kozai (2020); Hughes et al. (2026)
Compartmental Uncoupling (Soma vs. Axon)	Direct axonal recruitment bypasses somatic GABAA/GABAB inhibition that silences distal somata	False network silence when only somata are segmented, or neuropil bleed-through into silent somata	Axons active <40 μm; somata inhibited >60 μm	Eles & Kozai (2020); Suematsu et al. (2024)
Astrocytic Glutamate Clearance Failure	Stimulation >10 Hz impairs astrocytic GLT-1/EAAT2 uptake, leaving extracellular glutamate elevated	Lingering post-ICMS ΔF/F0 elevation driven by NMDA/mGluR influx rather than ongoing spiking	<20 μm from electrode; persists 10–20 s post-ICMS	Eles & Kozai (2020); Hughes & Kozai (2023)
Hemoglobin Optical Absorption ("Shadowing")	ICMS-evoked vasodilation increases local [HbO] and [HbT], absorbing 510–532 nm green emission	False neural suppression (uncorrected ΔF/F0 dips negative during and after stimulation)	Whole imaging field; lags onset by 1–2 s, persists >20 s post-ICMS	Suematsu et al. (2024)
Vascular Tissue Warping & Z-Drift	Dilating pial/penetrating arterioles mechanically push surrounding parenchyma laterally and axially	Artificial step-changes or dips in ΔF/F0 as somata shift out of fixed 2D ROIs and focal planes	5–15 μm tissue displacement around major vessels	Hattori & Komiyama (2022); Stringer et al. (2026)
Electrochemical Water Window & Shank Trauma	Exceeding −0.6 to +0.8 V (IrOx) generates gas microbubbles/ROS; rigid shanks sever local axons	Pathological Ca2+ waves in quiet cells, autofluorescent blebs, and optical shadowing under the probe	<50–100 μm from shank; acute to chronic	Lycke et al. (2023); Hughes et al. (2026)
Cold Objective Hypoxia & Inter-Trial Carry-Over	Unheated immersion objectives drop cortical temperature by 2–3◦C; short ITIs leave residual Ca2+	Reduced baseline pO2 and blood flow; inflated pre-stimulus baseline F0 compressing ΔF/F0	Superficial 100–300 μm; ITIs <60 s after high-charge trials	Podgorski & Ranganathan (2016); Dadarlat et al. (2024)

I. **Frequency, Amplitude, and ATP-Depleted Ca2+ Clearance Failure**
   A. High stimulation frequency and amplitude can trap calcium inside the cell because extrusion pumps run out of available ATP.
   B. Stalling PMCA and SERCA via ATP Exhaustion: Cytosolic Ca2+ clearance back to baseline (~50–100 nM) against a 10,000-fold gradient requires PMCA and SERCA pumps hydrolyzing 1 ATP per 1–2 Ca2+ extruded (Hughes et al., 2025). During continuous high-frequency (100–250 Hz) or high-amplitude (15–50 μA) ICMS, intense metabolic demand outpaces local capillary oxygen delivery within 10–20 s (Hughes & Kozai, 2023; Hughes et al., 2026). Depleted ATP stalls PMCA/SERCA clearance, trapping Ca2+ on GCaMP6s.
   C. Collapsing the Na+/Ca2+ Exchanger (NCX) Gradient: Under high firing loads, bulk extrusion also relies on NCX, powered by the Na+/K+-ATPase gradient (Hughes & Kozai, 2023). High-frequency ICMS simultaneously floods the cell with Na+ and depletes ATP; Na+/K+-ATPase fails, intracellular [Na+] accumulates, and NCX extrusion stalls or reverses into Ca2+-entry mode.
   D. Creating False "Steady-State" Fluorescence During Firing Failure: Because pulses directly depolarize neurons within 50–80 μm of the electrode, proximal somata show sustained, non-adapting ΔF/F0 plateaus (Hughes & Kozai, 2023; Hughes et al., 2025). However, simultaneous electrical recordings reveal that extracellular K+ buildup causes somatic depolarization block and axonal conduction failure (Eles & Kozai, 2020; Hughes et al., 2026). Because trapped Ca2+ cannot be extruded without ATP, 2P imaging reports continuous activation even after actual spiking stops.
   E. Inducing Long-Term Excitability Depression (SIDNE): Pushing metabolic and ionic homeostasis past recovery limits triggers stimulation-induced depression of neuronal excitability (SIDNE), elevating action potential thresholds up to 5× baseline and taking 2 to 17 days to recover (Hughes et al., 2026; Suematsu et al., 2024).

II. **Compartmental Uncoupling and Astrocytic Glutamate Spillover**
   A. Somatic GCaMP signals uncouple from axonal spiking and extracellular glutamate dynamics due to recruited perisomatic inhibition and impaired astrocytic uptake.
   B. Direct Axonal Firing Despite Somatic Inhibition: Threshold currents (5–15 μA) preferentially activate initial segments and passing axons rather than cell bodies (Eles & Kozai, 2020; Dadarlat et al., 2024). During a 30 s, 100 Hz train, recruited GABAA/GABAB perisomatic inhibition silences distal somata (>60 μm, showing 3× lower correlation with neuropil), while local axons within 20–40 μm continue spiking and releasing glutamate continuously (Eles & Kozai, 2020). Segmenting only somata falsely reports network silence.

C. Impaired Astrocytic Glutamate Clearance: High-frequency stimulation (>10 Hz) overwhelms astrocytic GLT-1/EAAT2 transporters, leaving extracellular glutamate within 20 µm of the electrode elevated for at least 10–20 s post-ICMS (Eles & Kozai, 2020; Hughes & Kozai, 2023). This lingering glutamate drives post-stimulus NMDA and mGluR/ER Ca2+ entry that mimics sustained action potential firing.

D. Subthreshold and Neuropil Contamination: Somatic GCaMP signals also capture subthreshold dendritic NMDA plateau potentials, back-propagating action potentials (bAPs), and dense surrounding neuropil fluorescence that bleeds through PSF tails into silent cell bodies (Chen et al., 2013; Grienberger et al., 2022).

III. Hemodynamic Optical Absorption and Vascular Tissue Warping

A. Stimulation-evoked vasodilation artificially depresses green fluorescence through hemoglobin absorption and shifts cell bodies out of focal masks.

B. Hemoglobin Absorption Shadowing: ICMS drives robust neurovascular coupling, triggering surges in total hemoglobin (Δ[HbT]) and oxyhemoglobin (Δ[HbO], 5× larger than Δ[HbR]) that lag stimulation onset by 1–2 s and persist for >20–24 s post-ICMS (Suematsu et al., 2024). Because hemoglobin strongly absorbs green emission (510–532 nm), dilating vessels above the imaging plane pull uncorrected ΔF/F0 downward, creating an artifactual suppression dip during late stimulation and recovery (Suematsu et al., 2024).

C. Non-Rigid Vascular Parenchymal Warping: Rapid arteriole dilation pushes surrounding cortical tissue laterally and axially by several micrometers (Hattori & Komiyama, 2022; Stringer et al., 2026). Fixed 2D ROIs lose overlap with drifting somata, generating artificial step-changes and transient dips in fluorescence that cannot be corrected by rigid image registration alone.

D. Cold Immersion Objective Hypoxia: Unheated water-immersion objectives cool superficial cortex (100–300 µm) by 2–3°C, reducing red blood cell velocity by ~33%, blood flow by ~22%, and baseline pO2 by ~20% (Podgorski & Ranganathan, 2016; CASRAI, 2026). This baseline hypoperfusion accelerates the onset of ICMS-induced hypoxia and ATP exhaustion.

IV. Electrode-Tissue Interface, Electrochemistry, and Optical Shadowing

A. Probe insertion trauma, electrochemical water-window exceedance, and physical shank occlusion alter local network excitability and optical access.

B. Insertion Trauma and Autofluorescent Blebs: Rigid probe insertion severs local horizontal axons, significantly reducing baseline correlation among neurons within 100 µm (Hughes et al., 2025), and creates bright, static autofluorescent debris that can be misclassified as active cells (Suematsu et al., 2024). Over chronic windows, reactive glial encapsulation (10–150 µm) increases interfacial impedance and distorts current spread (Eles & Kozai, 2020; Lycke et al., 2023).

C. Water-Window Exceedance and Gas Bubble Strain: Exceeding the electrochemical water window (-0.6 V to +0.8 V for IrOx) causes water electrolysis, generating ROS and gas microbubbles that mechanically strain adjacent membranes (Hughes et al., 2026; Lycke et al., 2023). This physical strain triggers pathological, non-physiological Ca2+ waves in silent cells while silencing active neurons.

D. Optical Shadowing by Opaque Shanks: Silicon or metal shanks cast an optical shadow cone that physically obstructs 920 nm excitation and 510 nm emission beneath the probe, degrading SNR at the stimulation site unless ultra-thin mesh arrays are used (Lycke et al., 2023).

V. Inter-Trial Interval Carry-Over

A. Insufficient recovery time between stimulation trains inflates baseline fluorescence and compounds metabolic exhaustion across trials.

B. Baseline F0 Inflation: High-amplitude (30–50 μA) or prolonged ICMS trains leave residual cytosolic Ca2+ and extracellular glutamate that require 10–20 s to clear (Eles & Kozai, 2020; Dadarlat et al., 2024). Inter-trial intervals (ITIs) shorter than 60 s cause subsequent trials to begin with an elevated baseline F0, mathematically compressing calculated ΔF/F0 responses.

C. Cumulative Metabolic and Excitability Fatigue: Rapid repetitive testing compounds ATP depletion, ion imbalance, and stimulation-induced depression of neuronal excitability (SIDNE), progressively elevating spiking thresholds and diminishing evoked calcium responses across successive blocks (Hughes & Kozai, 2023; Hughes et al., 2026).

Key Notes

I. Calcium Indicators:

A. GCaMP6s (Pan-neuronal): Used as a general calcium indicator to record overall neural activity in the visual cortex.
B. Thy1-GCamp6s (Excitatory): Utilized specifically in excitatory neurons.
C. AAV-Syn-GCaMP7b (Pan-neuronal): Expresses green fluorescence
D. AAV-FLEX-tdTomato (Inhibitory): Expresses red fluorescence specifically in inhibitory neurons to distinguish them from excitatory populations

II. Current Effects on Neuron Responses

A. Recruitment & Activation: Increasing stimulation amplitude recruits more excitatory and inhibitory neurons and yields larger evoked responses. The overall 3D volume of modulated cell bodies remains largely unchanged for excitatory neurons, but expands for inhibitory neurons with a moderate amplitude increase.
B. Excitation Types: As stimulation amplitude rises, indirect excitation increases while direct excitation decreases.
C. Inhibitory Sensitivity: Inhibitory neurons are preferentially activated at lower stimulation amplitudes due to their denser local axonal arborizations, which leads to early suppression of neural activity.
D. Response Stability: Higher stimulation amplitudes improve the stability of evoked responses by overriding and reducing dependence on ongoing, pre-stimulus neural activity in the brain.
E. Dynamic Amplitude: Evokes distinct onset and offset transients, reduces depression of neural calcium activity, and decreases total charge injection for sensory feedback in brain-computer interfaces by lower recruitment of neurons during long maintained periods of ICMS.

III. Frequency Effects on Neuron Responses

A. Activation Duration: High-frequency stimulation (e.g., 100-Hz) evokes the strongest initial neural activation during the first few seconds, but causes the activity to fade rapidly. Conversely, low-frequency patterns sustain activation effectively over longer durations.
B. Excitatory vs. Suppressive Dynamics: Low stimulation frequencies are broadly excitatory and correlate with the spatial extent of neural responses, whereas high frequencies are suppressive and promote the release of inhibitory neurotransmitters.
C. Dynamic Frequency: Evokes distinct onset and offset transients in a small subpopulation of neurons but also reduces depression in recruited neurons by reducing the rate of activation

D. Neuron-Specific Preferences:
    1. Inhibitory Neurons: Roughly 63% prefer Theta-burst stimulation (TBS), which uniquely avoids inducing post-stimulation depression in this population.
    2. Excitatory Neurons: Show a preference for 10-Hz burst patterns (34% preference). However, when subjected to TBS, excitatory neurons experience the least amount of post-stimulation depression compared to other patterns.
E. Post-Stimulation Depression: Higher-frequency patterns result in greater neural depression (activity falling below baseline) once the stimulation ends.

IV. Neural Plasticity & Behavioral Learning

A. Learning-Induced Adaptation: Behavioral training progressively lowers the detection threshold for intracortical microstimulation (ICMS) and increases the 3D volume of activation density at a fixed current.
B. Ensemble Stability: If the stimulation current is lowered to match the animal's newly improved detection threshold, the size and density of the activated neural ensemble remain stable, indicating a consistent network is required to maintain the behavioral response.
C. Pulse-Locked vs. Non-Pulse-Locked Neurons:
    1. Pulse-locked neurons: (Directly activated, primarily excitatory) exhibit increased excitability and reduced latency as learning progresses.
    2. Non-pulse-locked neurons: (Indirectly activated) do not change excitability but expand in population size. Their firing rates diverge significantly before movement initiation, directly predicting behavioral outcomes ("hits" versus "misses").

V. Computational Processing & Deconvolution (Suite2p & OASIS)

A. Algorithm Selection: Unsupervised non-negative deconvolution (NND) utilizing the OASIS algorithm is highly recommended over Convolutional Neural Networks (CNNs). NND runs orders of magnitude faster, resists overfitting, and avoids generating artifactual structures like spurious auto-correlations.
B. Data Pipeline Efficiency: The GPU-accelerated Suite2p pipeline isolates cellular compartments by their transient functional excursions above baseline using strict L0 sparsity constraints. This avoids falsely identifying neuropil or motion artifacts as active cells, outperforming older algorithms in speed and accuracy.

VI. Electrode Hardware & Safety (StimNETs)

A. Glial Scarring Prevention: Ultraflexible StimNETs (1.3 μm total thickness) prevent glial scarring, allowing the electrode contacts to remain intimately close to target neurons. This close proximity dramatically reduces activation thresholds to as low as 2 μA while improving stimulation focality and long-term stability.

B. Electrochemical Safety: These ultraflexible electrodes can safely deliver up to 50 μA of current while strictly operating within the water oxidation/reduction window (-0.6 V to 0.8 V). This constraint is critical for avoiding electrolysis and subsequent tissue damage.

VII. Volumetric Scanning via Electrically Tunable Lens (ETL)

A. Inertia-Free Axial Defocusing: Traditional stage- or objective-based axial stepping suffers from mechanical inertia, resonant vibration, and acoustic noise that can disturb awake animals and perturb delicate chronic implants such as ultraflexible StimNETs. Inserting an Optotune ETL into the excitation path allows Scanbox to modulate focal depth purely through electromagnetic shape alteration of an optical polymer fluid membrane.

B. Automated Calibration and Rapid Volume Definition: Scanbox streamlines volumetric imaging via the Volumetric Scanning GUI panel. Live navigation to the upper and lower boundaries latches "Top" and "Bottom" positions, automatically generating an evenly spaced lookup table across user-defined axial step sizes and dwell frame counts that uploads directly to the Scanbox acquisition tower FPGA.Telecentricity and Spatial Uniformity: Because changing focal depth via an ETL alters optical divergence before the objective entrance pupil, calibrated linearization lookup tables prevent field-of-view scaling distortions, enabling rapid volumetric stacks across hundreds of micrometers in tens of milliseconds per plane.

C. Real-Time Closed-Loop Depth StabilizationAxial Drift Susceptibility: While in-plane XY motion is routinely corrected online or offline, vertical Z-drift caused by animal locomotion, licking, thermal expansion, or slow tissue relaxation displaces cell bodies from thin two-photon optical sections, contaminating functional calcium Delta F/F traces with spurious transients.

D. Plugin Architecture: Scanbox exposes an extensible MATLAB callback architecture sbxplugin.m that intercepts incoming raw frame buffers synchronously during acquisition, executing real-time image processing without modifying the core acquisition engine.

E. 2D FFT Phase Cross-Correlation: By computing 2D fast Fourier transform (FFT) cross-correlations (search_fftalign.m) between live incoming frames and each slice of an a priori reference Z-stack, the depth stabilization plugin locates the global correlation peak to pinpoint the instantaneous focal depth.

F. Closed-Loop Hardware Correction: The calculated axial error offset is fed back directly to the Knobby motorized stage controller or the ETL driver, physically restoring the target focal plane dynamically throughout extended behavioral sessions.Automated Gain Control (AGC) of Excitation Laser PowerExponential Scattering Compensation: Two-photon excitation light scatters inhomogeneously in biological brain tissue according to the Beer-Lambert relation, where the

scattering mean free path in mouse neocortex is roughly 150–200 microns at 920 nm. Fixed laser power either saturates superficial layers or yields insufficient signal-to-noise ratios (SNR) at deeper cortical planes.

G. Pockels Cell Voltage Linearization: Laser power attenuation is mediated by an electro-optic modulator (EOM / Pockels cell). Because physical transmission follows a non-linear sinusoidal function of applied voltage, Scanbox calibrates and linearizes analog control voltages to enforce precise, predictable power delivery.
H. Dynamic Depth-Dependent Ramping: Scanbox integrates Automatic Gain Control directly with Z-stack execution. As the microscope steps axially, the acquisition engine computes the required compensation voltage, ramping excitation intensity exponentially with depth to maintain uniform photon emission and baseline fluorescence without inducing superficial thermal damage or phototoxicity.

Research Questions

I. **What causes neuron depression due to ICMS?**
   A. Recruited Inhibition: Depression during continuous stimulation (DCS) may be driven by the recruited inhibitory drive within the neural network, where ICMS activates inhibitory neurons that suppress the activity of excitatory neurons.
   B. Neuronal Exhaustion (Metabolic Stress): Continuous stimulation could cause cellular exhaustion or metabolic stress.
   C. Calcium Dynamics and ATP Depletion: A reduction in ATP availability induced by ICMS may limit a neuron's ability to pump out calcium Ca2+. This could result in stable calcium levels (and thus stable GCaMP fluorescence) even if the neuron's actual electrical excitability has decreased, complicating the interpretation of calcium imaging data.
   D. Depolarization Block: For depression across trials (SIDNE), accumulation of potassium ions in the extracellular space could cause a depolarization block, or electrochemical reactions at the electrode interface could produce toxic factors.

II. **How to differentiate between inhibitory and excitatory neurons?**
   A. Impose ICMS induced neural suppression to determine the states of activity after high amplitude, high frequency stimulation. Inhibitors should have higher levels of activity after high level ICMS, while excitatory neurons should have suppressed activity.
   B. Utilizing different calcium indicators: inputting a pan-neural calcium indicator with an excitatory or inhibitory specific indicator.

III. **How to find the same neurons across multi-day recordings?**
   A. PatchWarp integrated with Scanbox to move the microscope until achieving high accuracy at the same location.

IV. **What are the effects of dynamic amplitude and frequency on artificial sensation through ICMS?**

V. **How does the utilization of calcium indicators (like GCaMP6s) affect in vivo 2 photon microscopy studies of mice?**
   A. Could the decay rate of the fluorescence and speed of activation affect our visual indication of neuron spiking?
   B. Could it hide neural suppression due to ICMS due to decay rate causing us to view higher activity rates than what's actually occurring. Does it use more ATP to activate it thus requiring a larger stimulation amplitude in neurons to cause firing, or could we have non-fluorescent firing in neurons which don't activate the fluorescence?